

4.1 Answer each of the following:

- a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- c. Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2 .

4.1

a. Recall that $SST = \sum_{i=1}^N (y_i - \bar{y})^2 = \sum_{i=1}^N y_i^2 - N \bar{y}^2$.

So

$$R^2 = \frac{SSR}{SST} = \frac{375.47}{7825.94 - 20 \cdot 19.21^2} \approx \underline{0.8429}$$

b. Recall that in a simple regression, $R^2 = 1 - \frac{SSE}{SST}$

So $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{SSE}{N-2} = \frac{SST(1-R^2)}{N-2} \approx \underline{8.4249}$

c. In a simple regression, we have

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum \hat{e}_i^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{182.85}{631.63} = \underline{0.7105}$$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- a. the predicted value of y for $x_0 = 4$.
- b. the $se(f)$ corresponding to part (a).
- c. a 95% prediction interval for y given $x_0 = 4$.
- d. a 99% prediction interval for y given $x_0 = 4$.
- e. a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

4.3

a. The predicted value of y for $x_0 = 4$ is $\hat{y}_0 = 1.2 + 0.8 \cdot 4 = \underline{4.4}$

b.
$$\begin{aligned} \hat{\text{var}}(f) &= \hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right] \\ &= \frac{3.6}{3} \left[1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right] \\ &= \underline{2.52} \end{aligned}$$

So $se(f) = \sqrt{\hat{\text{var}}(f)} \approx \underline{1.5875}$

c. The 95% prediction interval for y given $x_0 = 4$ is

$$\begin{aligned} \hat{y}_0 \pm t_{3;0.025} \cdot se(f) &= 4.4 \pm 3.182 \cdot 1.5875 \\ &= \underline{4.4 \pm 5.051} \quad \text{or} \quad \underline{[-0.651, 9.451]} \end{aligned}$$